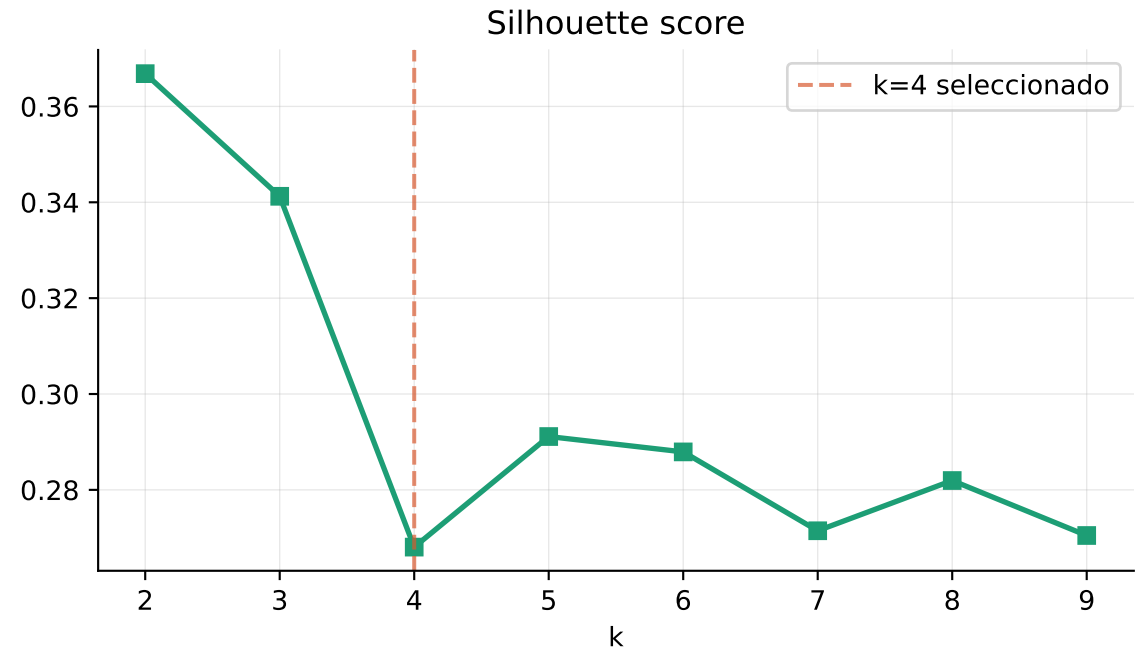
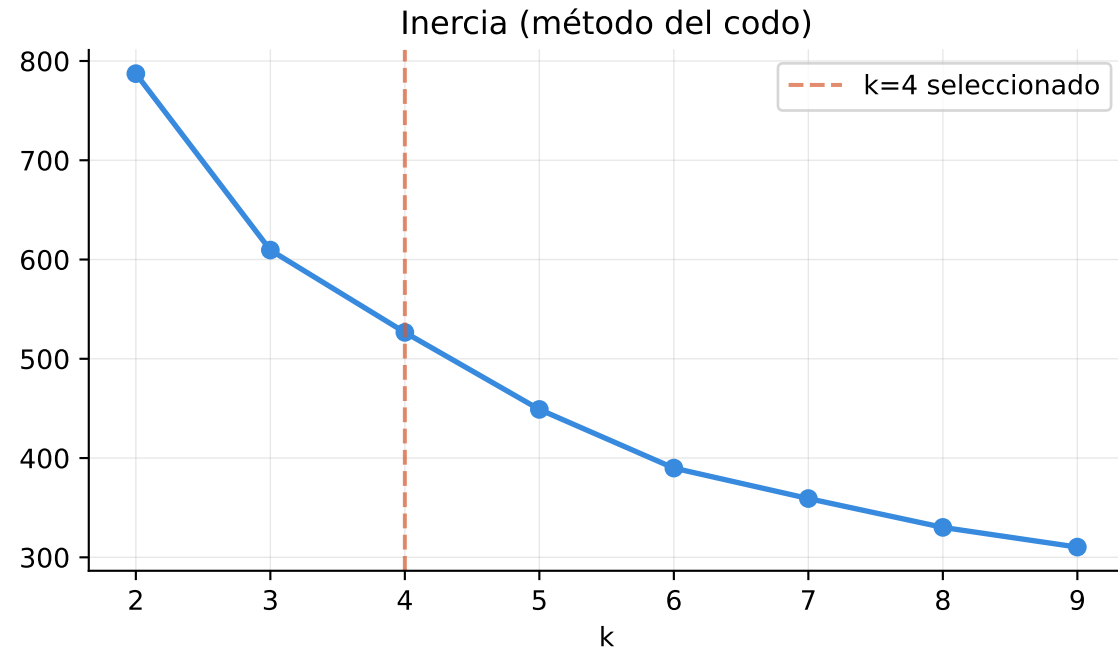
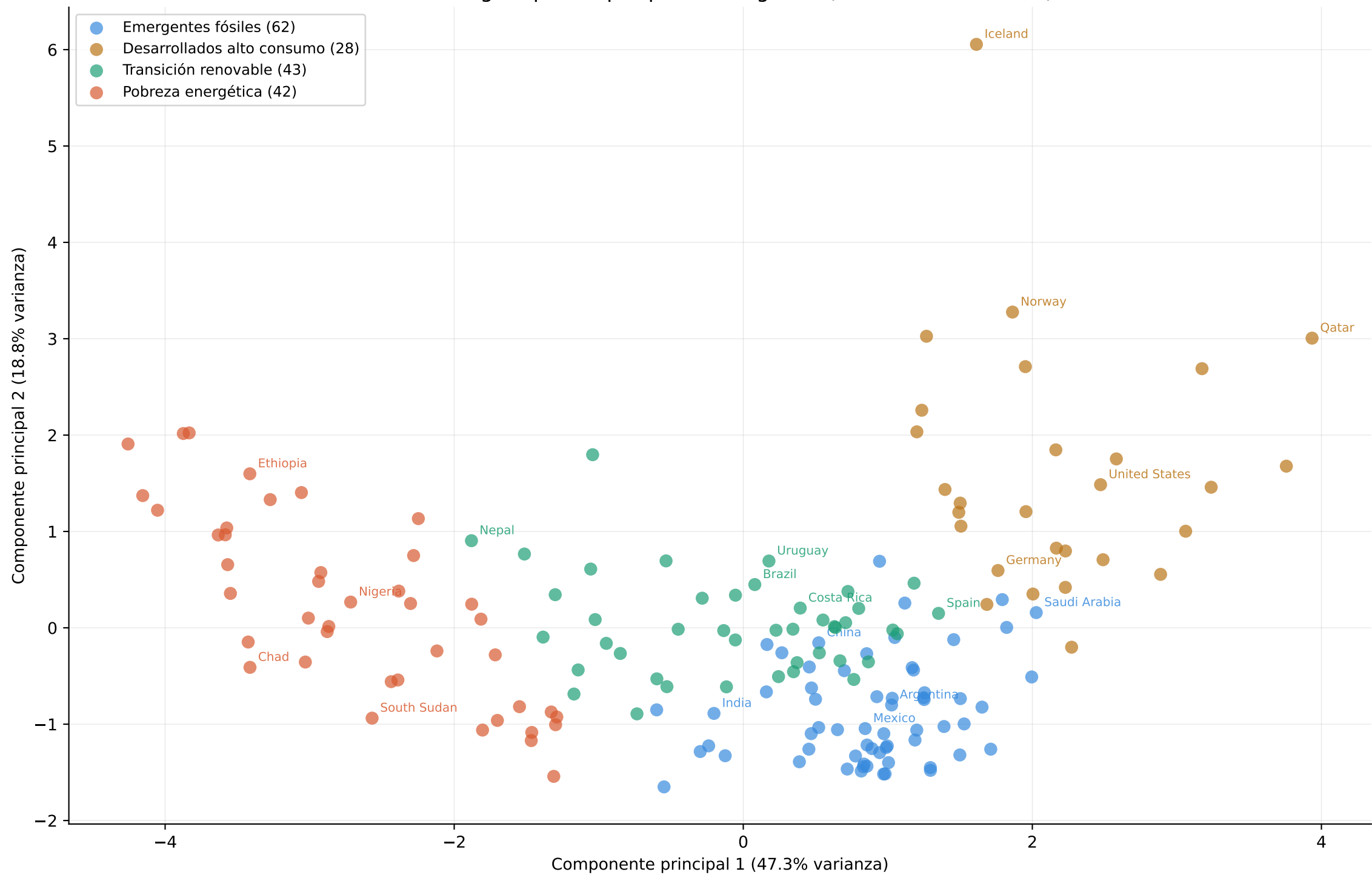


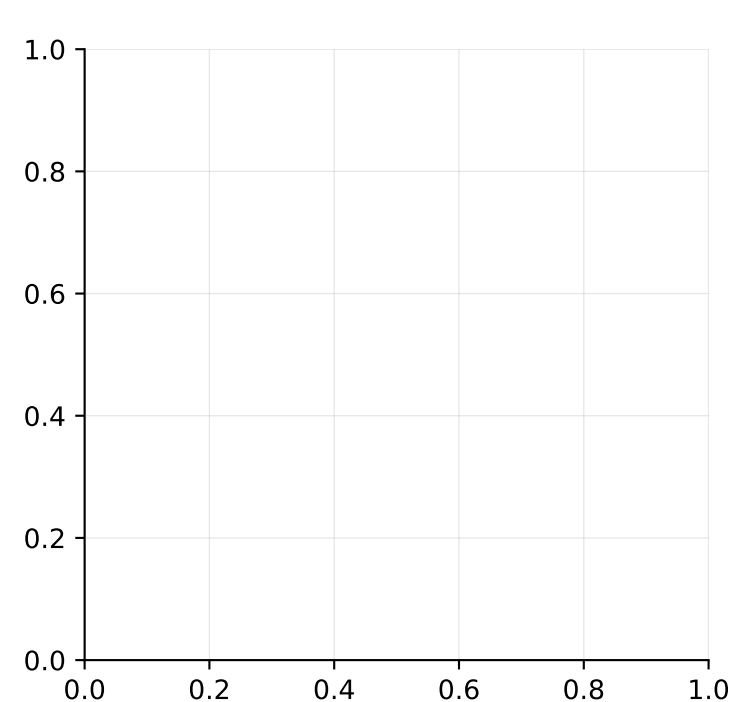
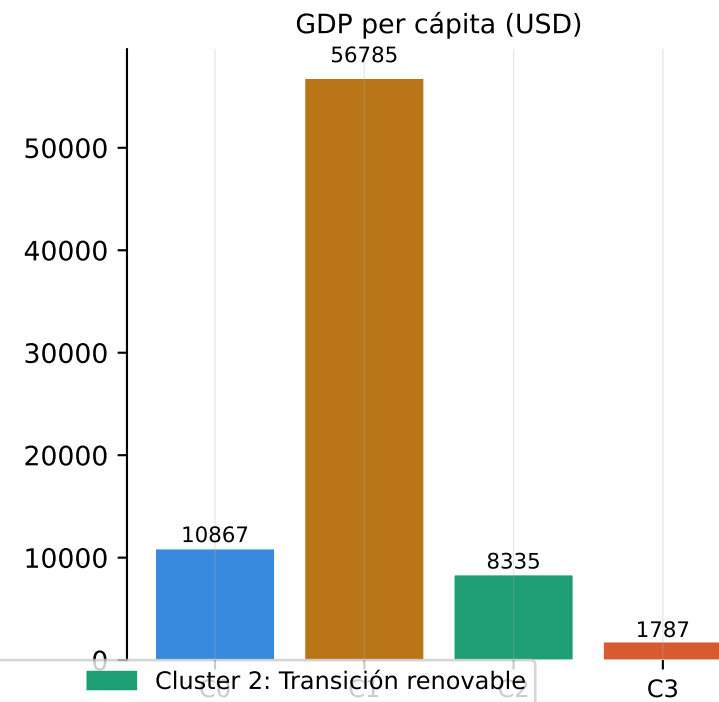
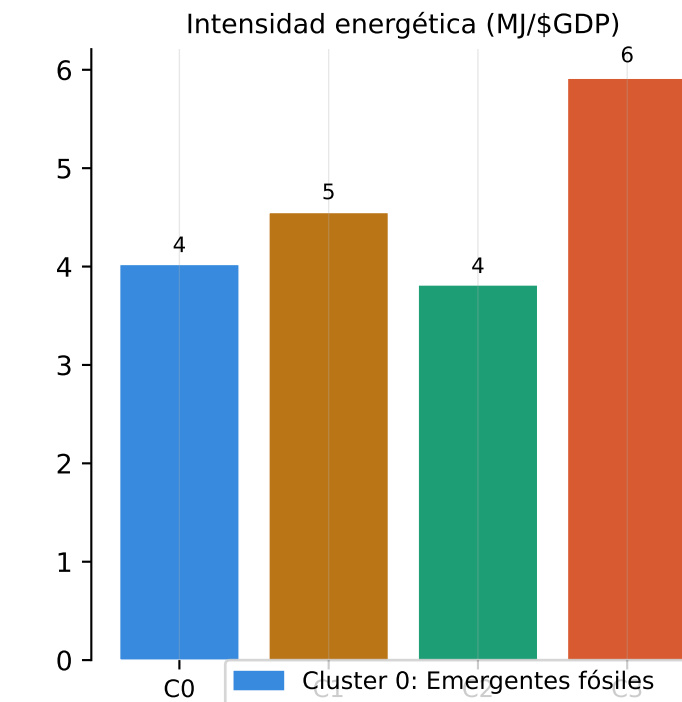
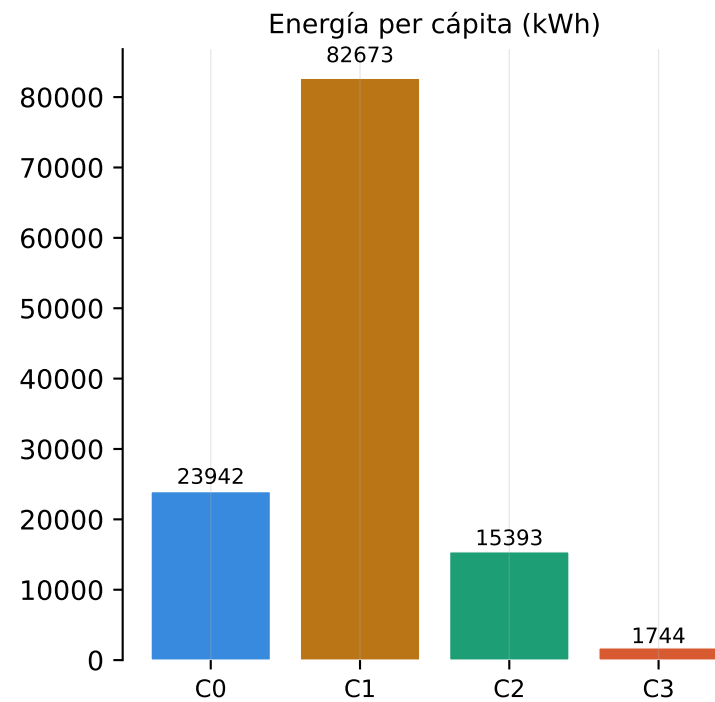
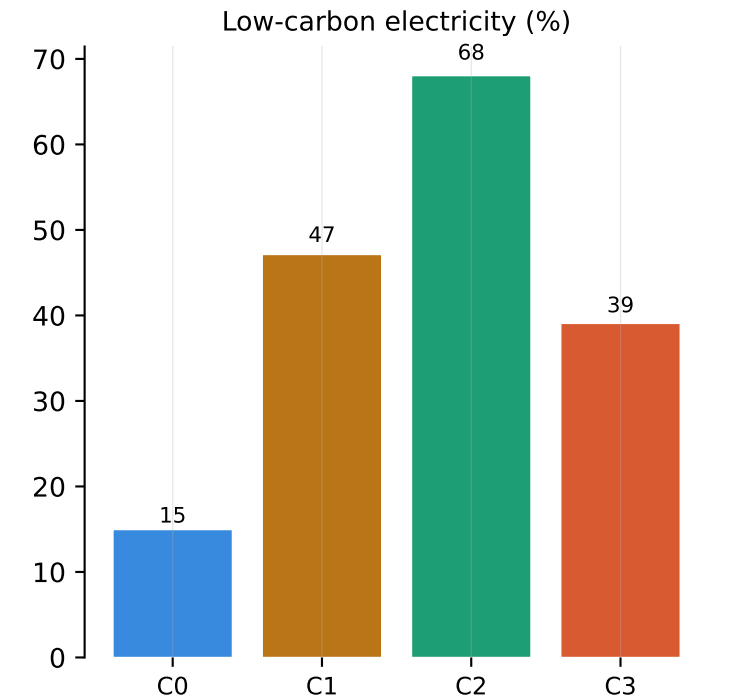
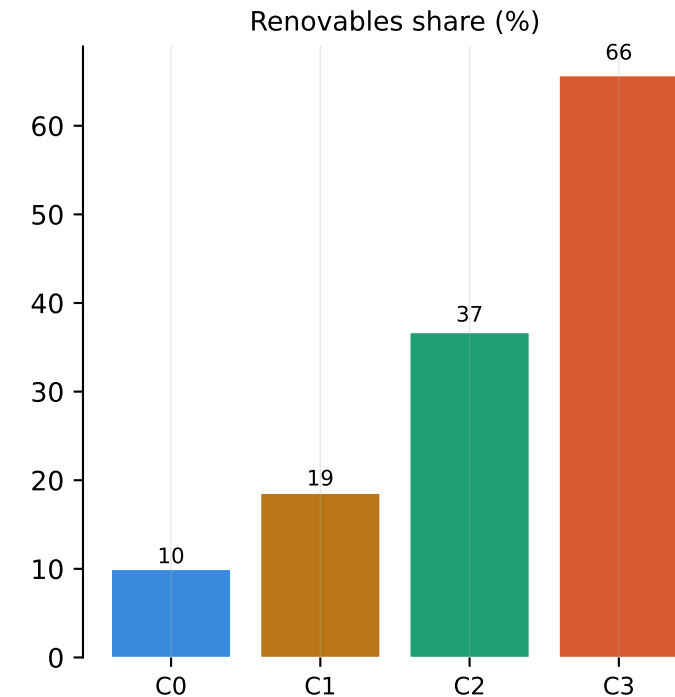
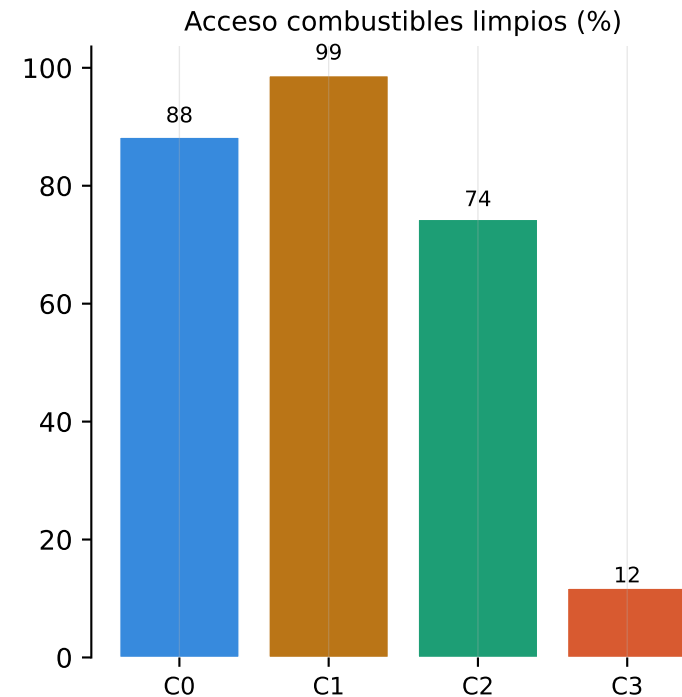
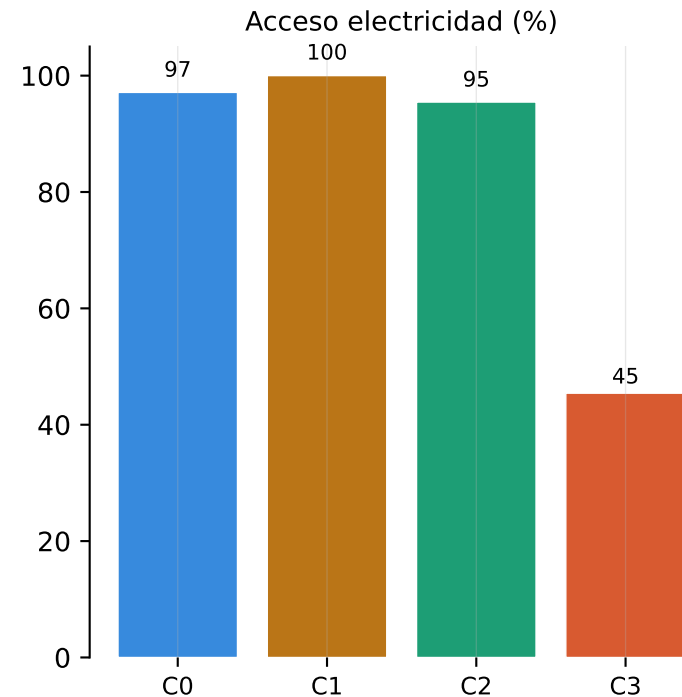
Selección de k óptimo



Clustering de países por perfil energético (PCA 2D, datos 2019)



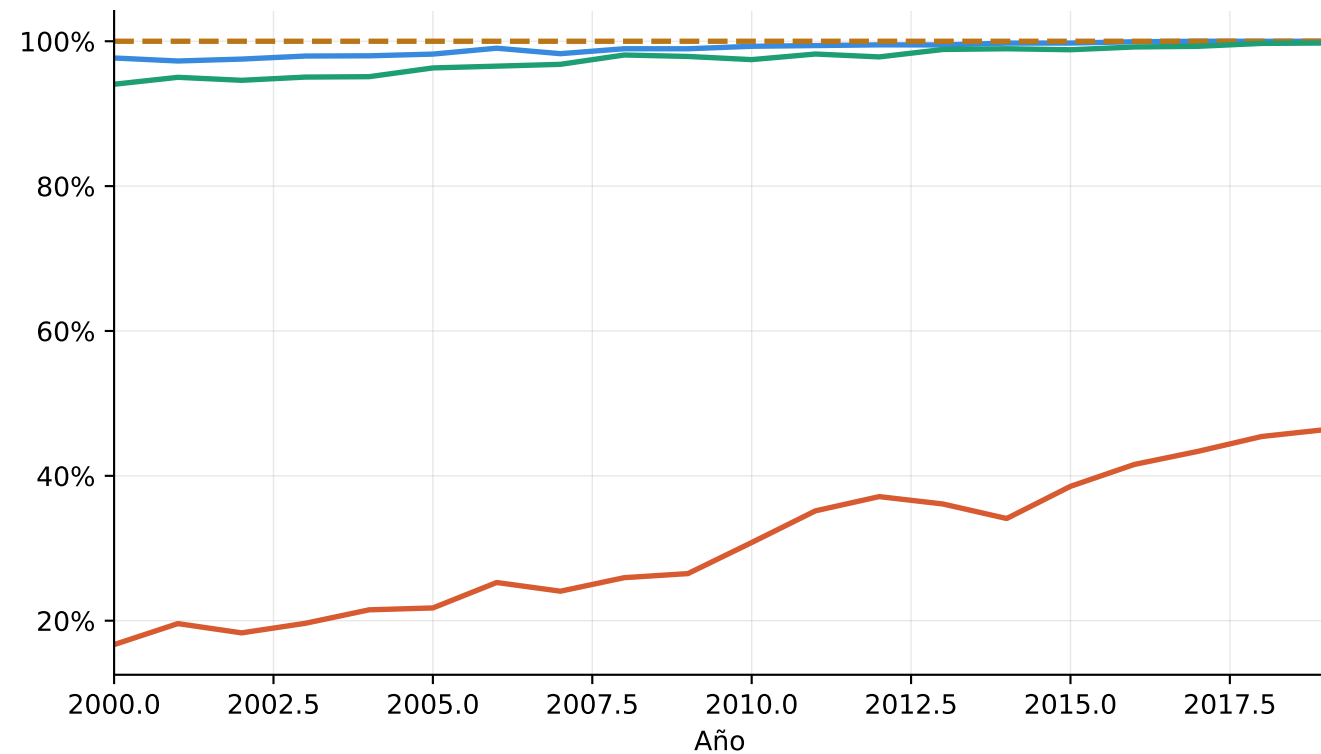
Perfil promedio por cluster — variables originales



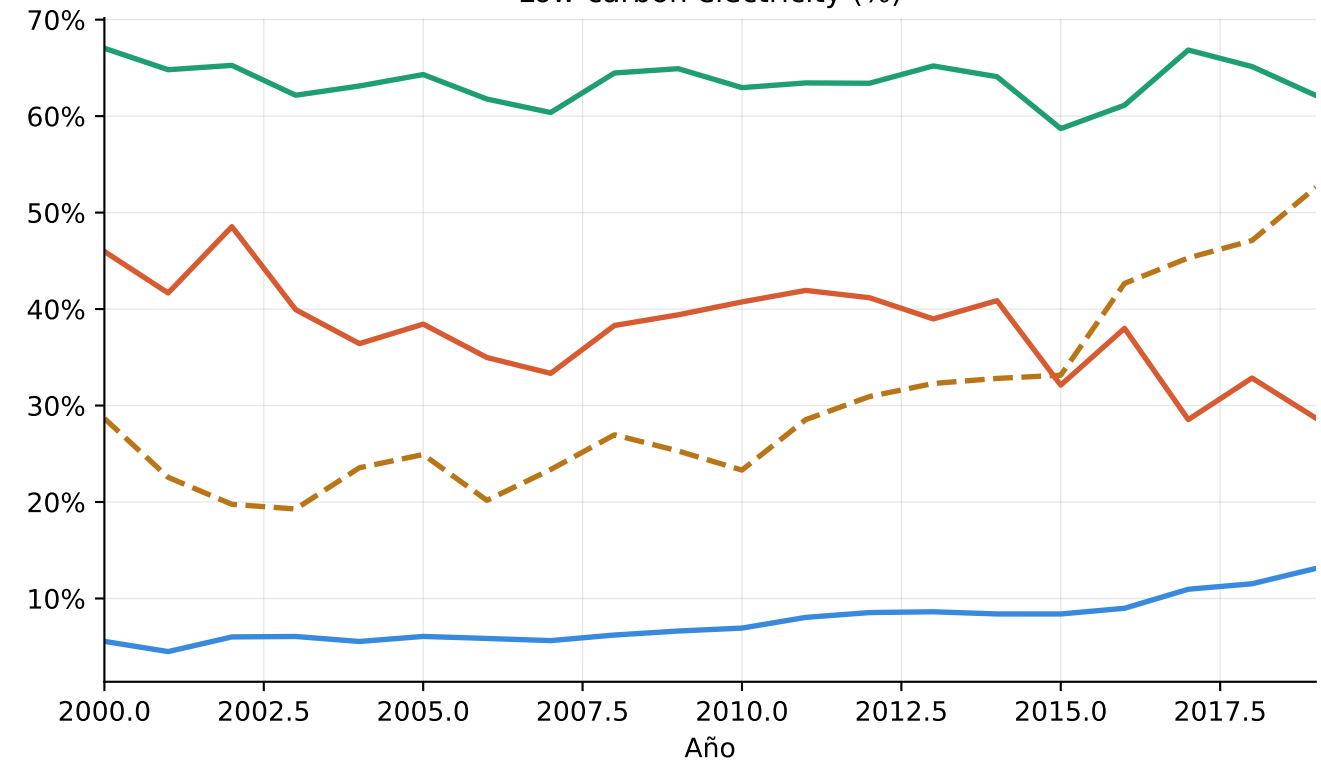
Cluster 0: Emergentes fósiles Cluster 1: Desarrollados alto consumo Cluster 2: Transición renovable Cluster 3: Pobreza energética

Evolución de indicadores por cluster (mediana, 2000-2019)

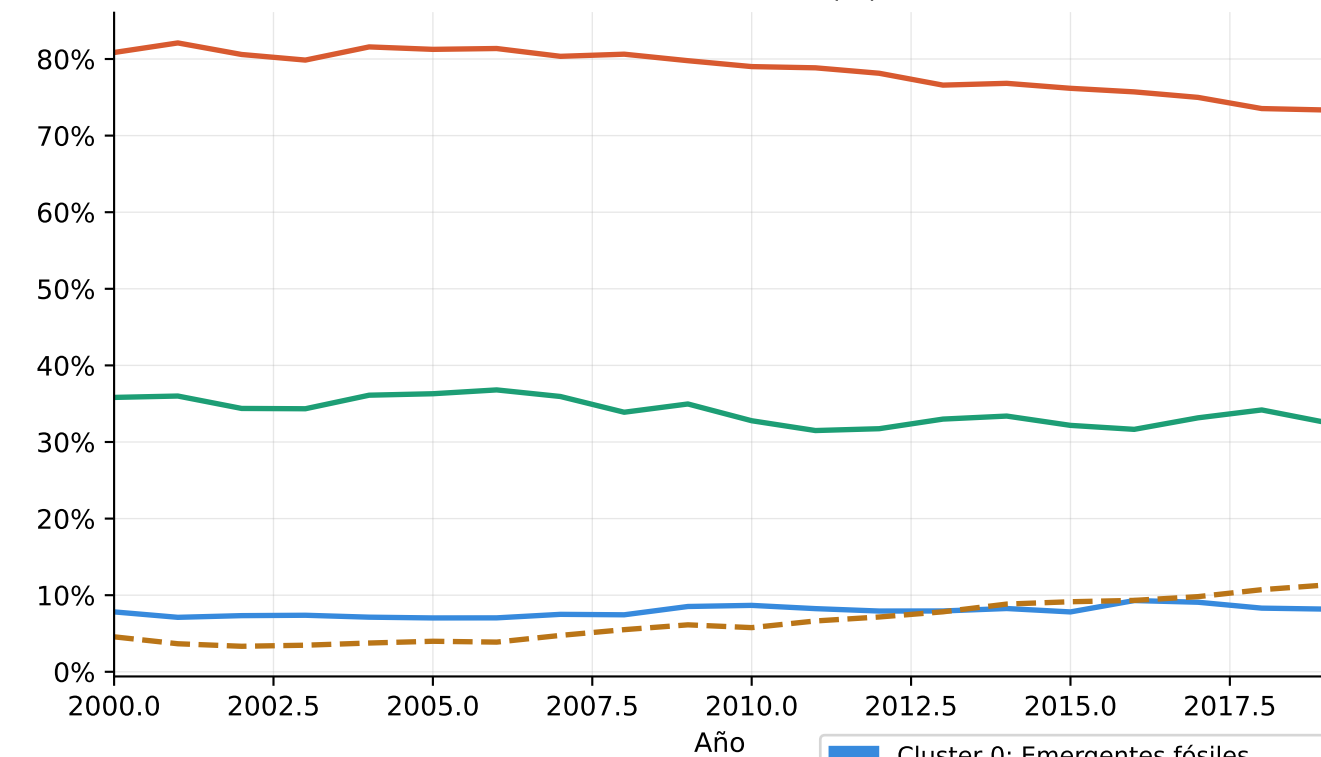
Acceso a electricidad (%)



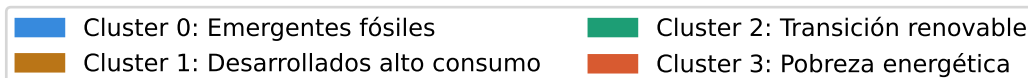
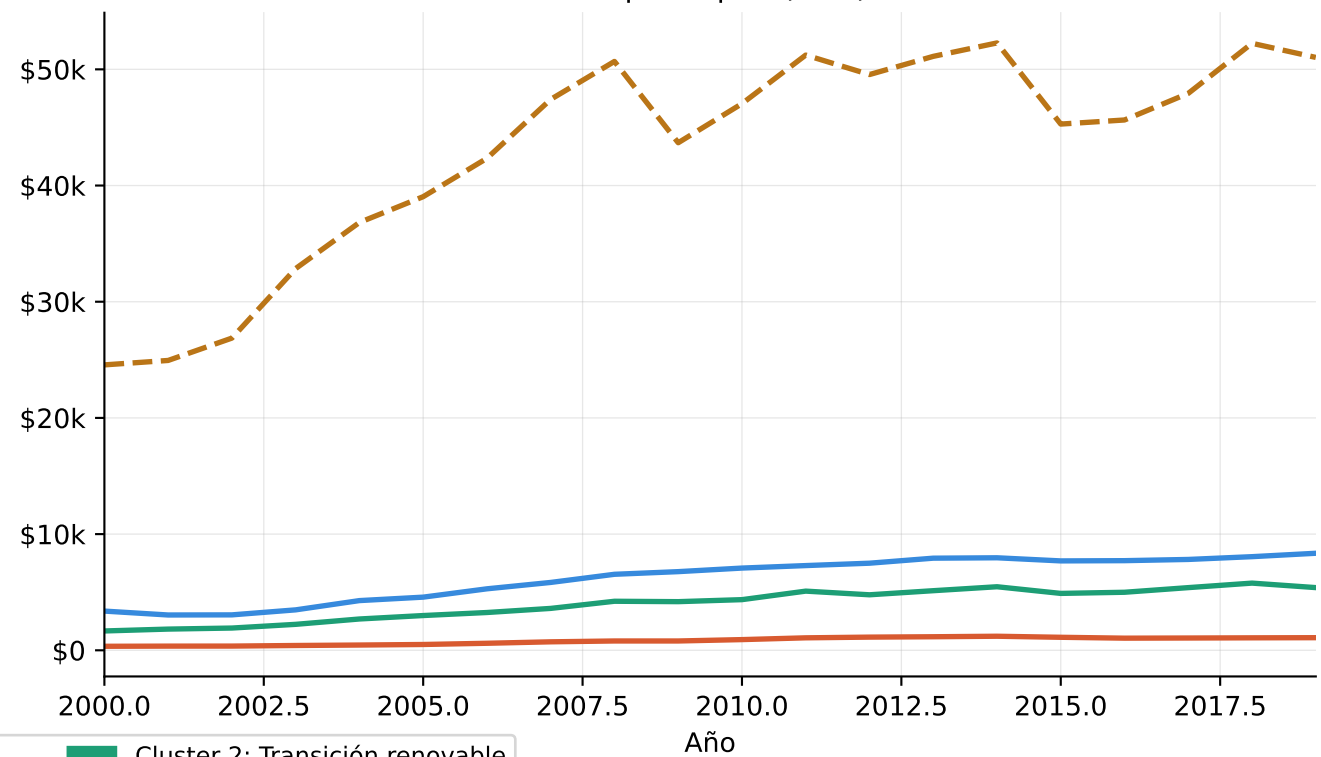
Low-carbon electricity (%)



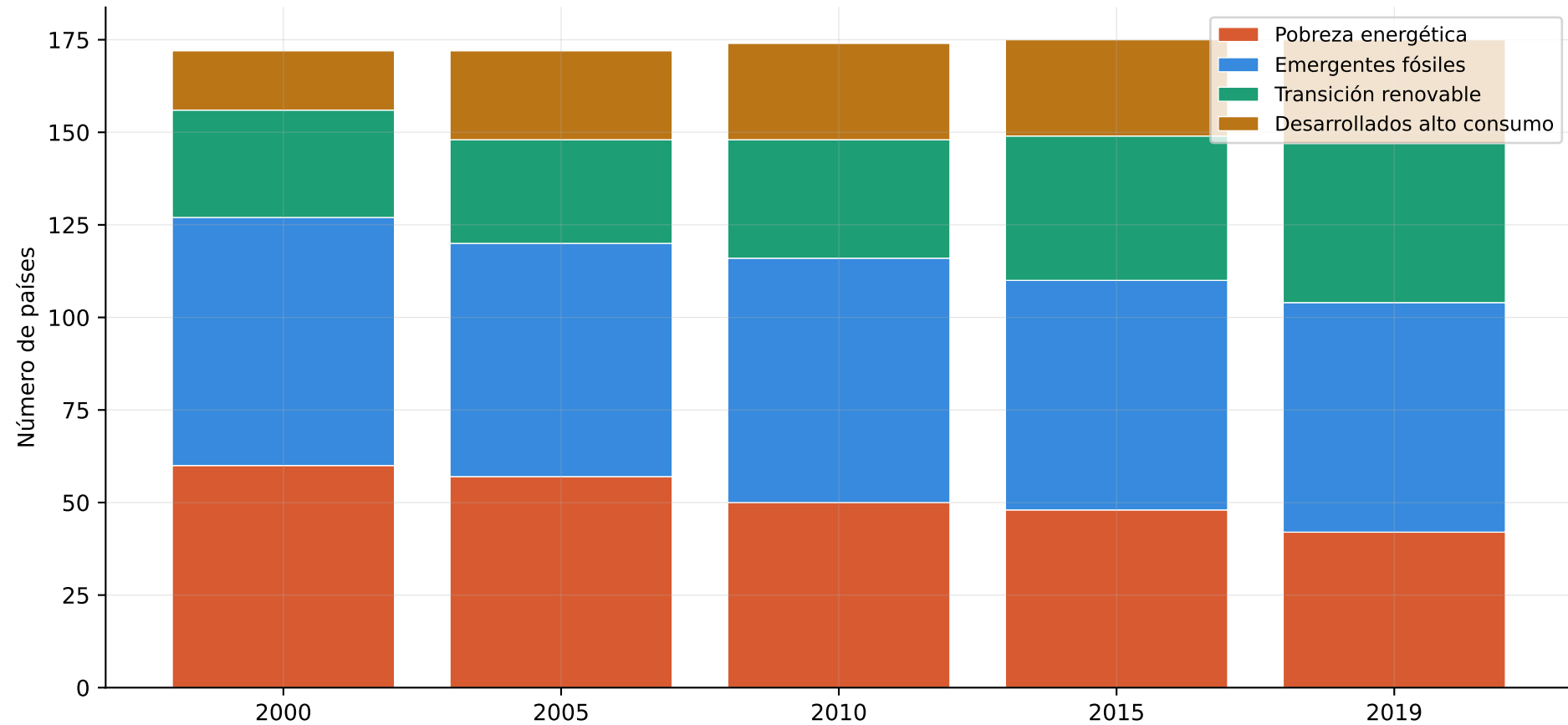
Renovables share (%)



GDP per cápita (USD)



Distribución de países por cluster a lo largo del tiempo



Matriz de migraciones entre clusters (2000 → 2019)

